$f : \mathbb{Z} \longmapsto \{0,1\}$

binary operator is ~~[illegible]~~

$f(\mathbb{Z} \setminus \{0\}) = 1$

$f(0) = 0$

```
neg     rax
sbb     rax, rax
neg     rax
```


neg rax

↳ {2's complement negation} of RAX

↳ [rax] == 0 → CF = 0 ; ELSE → CF == 1 ie op. performed

sbb rax, rax ~ "subtract w/ borrow / carry"

↳ dest = dest − src (IF CF=1) is = 0 (−1) ?

neg rax

$f(\mathbb{Z} \setminus \{0\}) = 1$

$f(\{0\}) = 0$

ie $f \xrightarrow[\text{fn}]{\text{binary}} \{0,1\}$

rax := bool(rax)

0 → 0 ⇒ CF = 0

ex | rax := 0x00 ~ 0000 0000

neg rax ie ~rax = 1111 1111

~rax + 1 = 0xff + 0x01

= 0x~~01~~00

= 0x00

sbb rax, rax ie rax = 0 − 0 ~~= 1 (CF ≠ 0)~~

= 0

neg rax CF = 0

ex | rax := 0x~~ff~~ ~ 0000 1111

0x0F neg rax ⇒ rax = 1111 0000 + 1

↓ CF == 1 = 1111 0001

= $-2^7 + 2^6 + 2^5 + 2^4 + 2^0$

0x01 = −64 − 32 − 16 − 15 = −128 + 64 + 32 + 16 + 1

sbb rax = 1101 0000 − 1110000 = 0000 0000 − 1

= −1 = 1111 1111

neg rax ⇒ 0000 0000 + 1 = 0000 0001